2313666 殷家兴 自动化 运控③

1. 解：(1)

$$\alpha = \frac{U_{nm}^*}{n_N} = 0.01\ (V\cdot min)/r \qquad \beta = \frac{U_{im}^*}{I_N} = 0.75\ V/A$$

当系统稳定在 ~~$U_{im}^* = 15V$~~ $U_n^* = 5V$ 时

$$n = \frac{U_n^*}{\alpha} = 500\ r/min$$

$$U_n = U_n^* = 5V$$

$$U_i^* = U_i = \beta I_{dL} = 7.5V \qquad I_d = I_{dL} = 10A$$

$$U_c = \frac{C_e n + I_d R}{K_s} = \frac{0.127 \times 500 + 10 \times 2}{20} V = 4.175V$$

(2) 当电动机负载过大而堵转时

$$U_i^* = U_{im}^* = 15V$$

$$I_d = 2I_N = 40A$$

$$U_c = \frac{U_{d0}}{K_s} = \frac{I_d R}{K_s} = 4V$$

2. 解：$\beta = \dfrac{U_{im}^*}{I_{dm}} = 0.1\ V/A$

(1) $I_{d1} = 40A$ 时 $\quad U_{i1}^* = \beta I_{d1} = 4V$

$I_{d2} = 70A$ 时 $\quad U_{i2}^* = \beta I_{d2} = 7V$

$$\Delta U_i^* = U_{i2}^* - U_{i1}^* = 3V$$

U_i^* 增大 3V

(2) ~~$U_c = \frac{U_{d0}}{K_s} = \frac{E + I_d R}{K_s}$~~ ~~从主向~~ 沿前向通路正向看，$U_c$ 随 U_i^* 变化而变化，U_c 与 U_i^* 成正比，设比例系数为 K_c，则 $\Delta U_c =$ ~~$3V\cdot K_s$~~ $3K_s V$

~~$\Delta U_c =$~~

(3) ~~$U_c =$~~ 当双闭环调速系统处于稳态时，有关系

$U_c = \dfrac{U_{d0}}{K_s} = \dfrac{C_e n + I_d R}{K_s}$ 成立，由该式可以看出，

ACR 的输出量控制电压 U_c 的大小与 n 和 I_d 有关

3. 解：采用PI调节器

$$W_{\cancel{ASR}}(s) = \frac{K_n(\tau_n s+1)}{\tau_n s}$$

$$W_n(s) = W_{\cancel{ASR}}(s) \cdot W_{obj}(s)$$

$$= \frac{10K_n(\tau_n s+1)}{\tau_n s^2(0.02s+1)}$$

查表可知，若满足阶跃输入下 $\sigma\% \le 28\%$、$t_s \le 0.26s$，则应选取 $h=8$

则 $\tau_n = hT = 8 \times 0.02 = 0.16$

$$\frac{10K_n}{\tau_n} = \omega_1\omega_c = \frac{h+1}{2h^2T^2}$$

$$K_n = \frac{45}{16} = 2.8125$$

故调节器 $W(s) = \dfrac{2.8125(0.16s+1)}{0.16s}$